Assesment-5 Ca

Aim: To write verilog code for SR, JK, D, T, flip flop & verify it functionally

Tools Required: Vivado

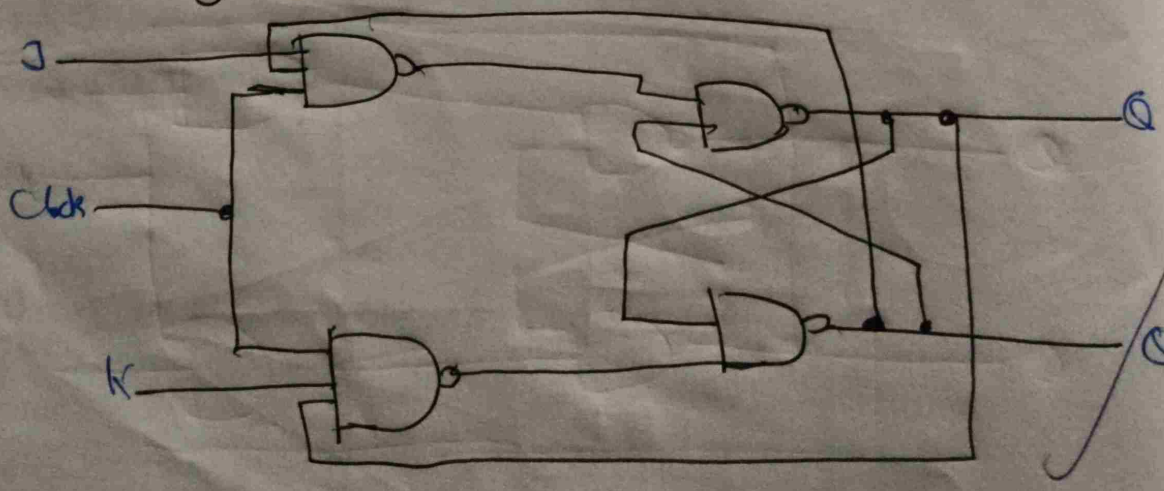
SR FLIP

@Theory: a two-state memory circuit with set(S) & Reset(R) inputs that stores a single bit of info.

Truth table:

S	R	Q(n+1)	State
0	0	Qn	No change
0	1	0	RESET
1	0	1	SET
1	1	X	Invalid

Circuit Diagram

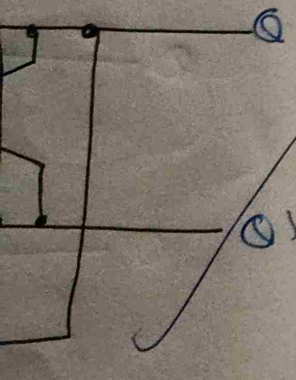


5 (a)

JK, D, T, flip flops

cuit with set (S)
tores a single

tate
o change
ESET
SET
alid



Code

```
module sr_ff (clk, rst, s, r, q);
input clk, rst, s, r;
output reg q;
always @(posedge clk or posedge rst)
begin
if (rst == 1)
q <= 0;
else if (s == 0 && r == 0)
q <= q;
else if (s == 0 && r == 1)
q <= 0;
else if (s == 1 && r == 0)
q <= 1;
else if (s == 1 && r == 1)
q <= 1'bx;
end
endmodule

module tb();
reg clk, rst, s, r;
wire q;
sr_ff uut (clk, rst, s, q, r);
always begin
#10 clk = ~clk;
end
initia
```

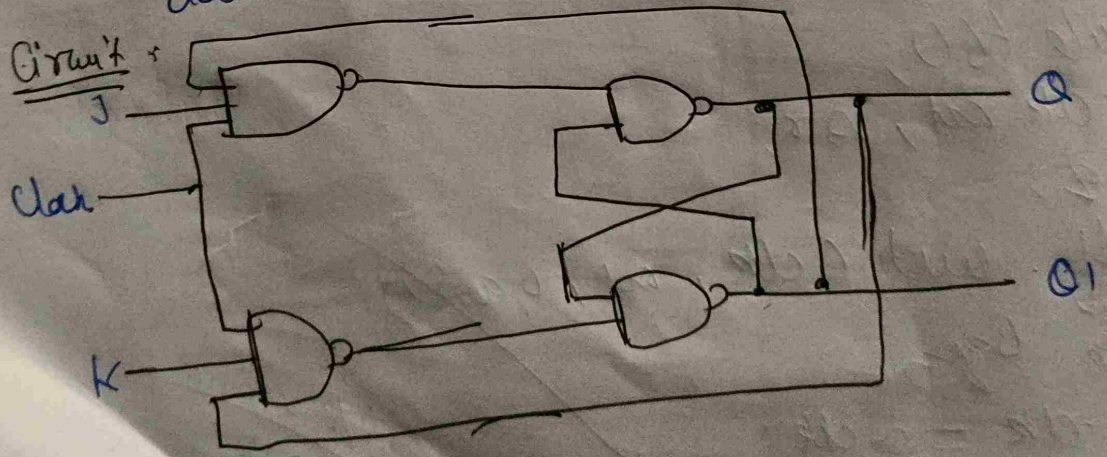
O/p verified
25-09-2025

```
begin
  clk =0; rst =0; r=0; s=1;
#5 rst =1;
#5 rst =0; s=2; r=0;
#20 s=1; r=0;
#20 s=1; r=1;
#20 s=0; r=0;
#20 rst =1;
#5 rst=0;
$ stop;
end
endmodule
```

JK_Flip

theory: a type of electronic memory device that stores a single bit of information and can be use to manipulate digital data.

Circuit:



JK

Truth table

J	K	Q(n+1)	State
0	0	Qn	No change
0	1	0	RESET
1	0	1	SET
1	1	Q n'	TOGGLE

Code

```
module jk_ff (clk, rst, j, k, q);
input clk, rst, j, k;
output reg q;
always @ (posedge clk or posedg rst)
begin
if (rst == 1)
q <= 0;
else if (j == 0 && k == 0)
q <= q;
else if (j == 1 && k == 0)
q <= 1;
else if (j == 1 && k == 1)
q <= ~q;
end
endmodule
```

*/p verified
25-09-2025

```
module tb();
reg clk, rst, j, k;
wire q;
jk_ff ut (clk, rst, j, k, q);
always begin
# clk = ~clk;
end
initial
begin
clk = 0; rst = 0; j = 0; k = 0;
# 5 rst = 1;
# 5 rst = 0; j = 0; k = 0
# 25 j = 0; k = 1;
# 25 j = 1; k = 1;
# 25 j = 1; k = 0;
# 5 rst = 1;
# 25 rst = 0; j = 1; k = 1;
$stop;
end
endmodule
```

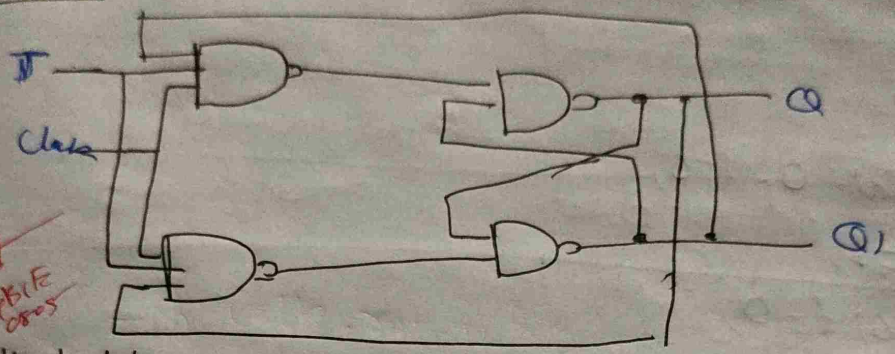
theory:

Circuit

T flip flop

theory: a digital electronic circuit with a single input (T) that changes its output state (Q) with every clock pulse with T is high, and maintains its current state when T is low.

Circuit



Truth table

T	Q(n+1)	State
0	Q	No change
1	Q1	Toggle

Code

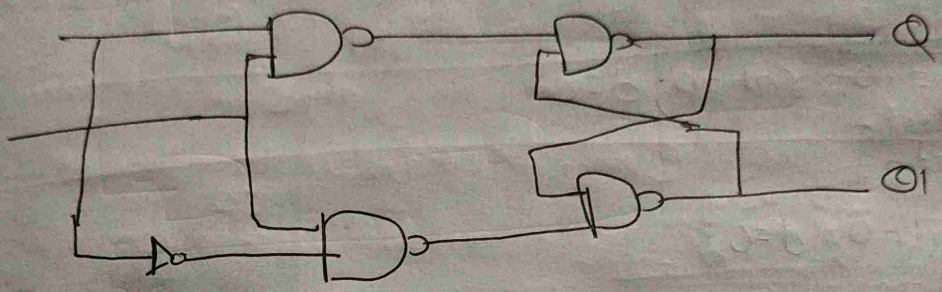
```
module t_ff (clk, rst, t, q);
input clk, rst, t;
output reg q;
always @(posedge clk or posedge rst)
begin
if (rst == 1)
q <= 0;
else if (t == 0)
q <= q;
else
q <= ~q;
end
```

```
module tb();
reg clk, rst, t;
wire q;
T_ff uut (clk, rst, t, q);
always begin
#10 clk = ~clk;
end
initial
begin
clk=0; rst=0; t=0;
#5 rst=1;
#5 rst=0; t=0;
#15 t=1;
#10 t=0;
#15 t=1;
#5 rst=1;
#5 rst=0; t=1;
$stop;
end
endmodule
```

D flip flop

Theory: a clock sequential logic circuit that stores a single bit of binary data, acting as a delay element by copying its data (D) input to its output (Q) at the moment of a clock edge.

Circuit:



Truth-Table

D	Q(n+1)	State
0	0	RESET
1	1	SET

Code:

```
module d_ff(clk, rst, d, q);
input clk, rst, d;
output reg q;
always @(posedge clk or posedge rst)
 begin
if(rst==1)
q<=0;
else
q<=d;
end endmodule
```

O/P verified
Date
25-09-25

```
module tb();
reg clk, rst, d;
wire q;
d_ff uut (clk, rst, d, q);
always begin
#10 clk = ~clk;
end
initial
begin
clk = 0, rst = 0; d = 0
#5 rst = 1;
#5 rst = 0; d = 0;
#15 d = 1;
#10 d = 0;
#15 d = 1;
#5 rst = 1;
#5 rst = 0; d = 1;
$stop;
end
endmodule
```

Aim:

Tod

The

Assesment - 5(b)

Aim: To write the verilog code for SISO shift register

Todo Requird: VSIM

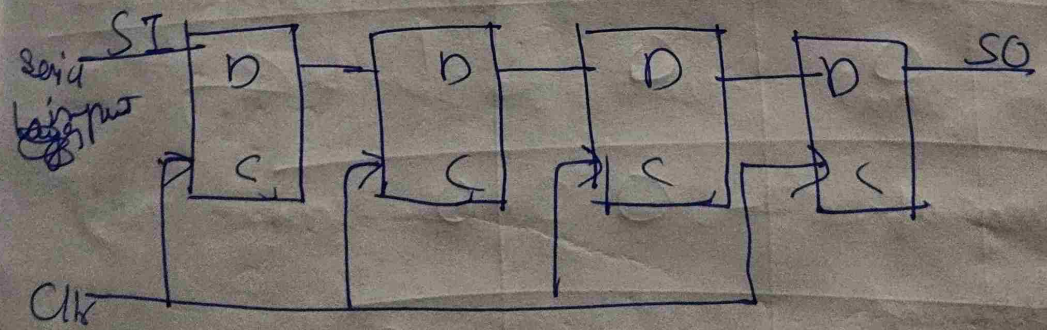
Theory:

SISO:- fundamental sequential digital logic circuits that store & shift data bit-by-bit, controlled by a clock signal

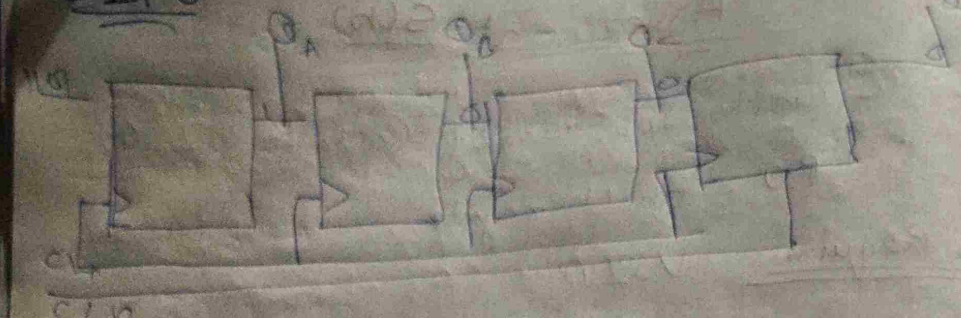
SIPO: a shift register also uses a series of interconnected flip-flops, but each flip-flop's output is accessed as a separate parallel output

Circuit Diagram

SISO



SIPO



Truth table:

SISO.

CLK	'Q3'	'Q2'	'Q1'	'Q0'
Initially	0	0	0	0
1st falling edge	1	0	0	0
2nd	1	1	0	0
3rd	1	1	1	0
4th	1	1	1	1

SIPO

Clock Pulse	Q1	Q2	Q3	Q4
0	1	0	0	1
1	1	1	0	0
2	0	1	1	0
3	0	0	1	1

Code

```
module SISO (so, si, clk, rst);
output so;
input si, clk, rst;
reg [3:0] q;
always @ (posedge clk or posedge rst)
begin
if (rst)
q <= 4'b0;
else
q <= {si, q[3:1]};
end
assign s = q[0]
endmodule
```

21BCE0505

```
module siso_test ();
wire so;
reg si, clk, rst;
siso
s1 (so, si, clk, rst);
always
# 5 clk = ~clk;
initial begin
si = 0; clk = 0; rst = 1;
# 10 rst = 0;
# si = 1;
```

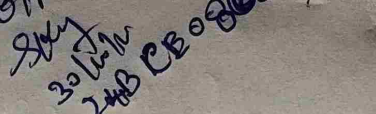
#10 si=0;
#10 si=0;
#10 si=1;
#10 si=1;
#50 $stop;
end
endmodule

Code-

```
module sipo(q,si,clk,rst);
output [3:0]q;
input si,clk,rst;
reg [3:0]q;
always @(posedge clk or posedge rst)
begin
if(rst)
q<=4'b0;
else
q<={si,q[3:1]};
end
endmodule

module sipo_tb;
wire [3:0]q;
reg si,clk,rst;
sipo dl(q,si,clk,rst);
always
#5 clk=~clk;
```

```
initial
  begin
  a = 0;
  clk = 0;
  rst = 1;
  #10 rst = 0;
  #10 a = 1;
  #10 a = 0;
  #10 a = 0;
  #10 a = 1;
  #10 a = 1;
  #10 a = 1;
  #50 $stop;
  end
endmodule
```

o/p



Conclusion

The Verilog code is successfully verified

Assesment - 15 (c)

Aim:- To verify verilog code for (a) up-down counter
(b) Modulo N counter
(c) Johnson counter

Tools Required: Modelsim

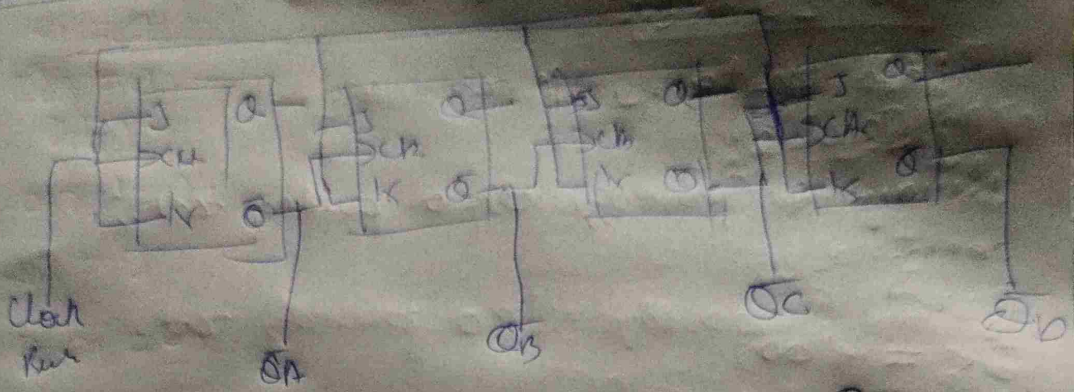
Theory:- Up-down Counter: A counter that can be program to count in either an ascending or descending order based on a control signal.
It increases from a low value to a high value or decreases from a high value to a low value.

Modulo-n-Counter: A counter with a specific number of unique states, denoted by 'n'. It will count from 0 to n − 1 & then reset, having a total of n unique state.

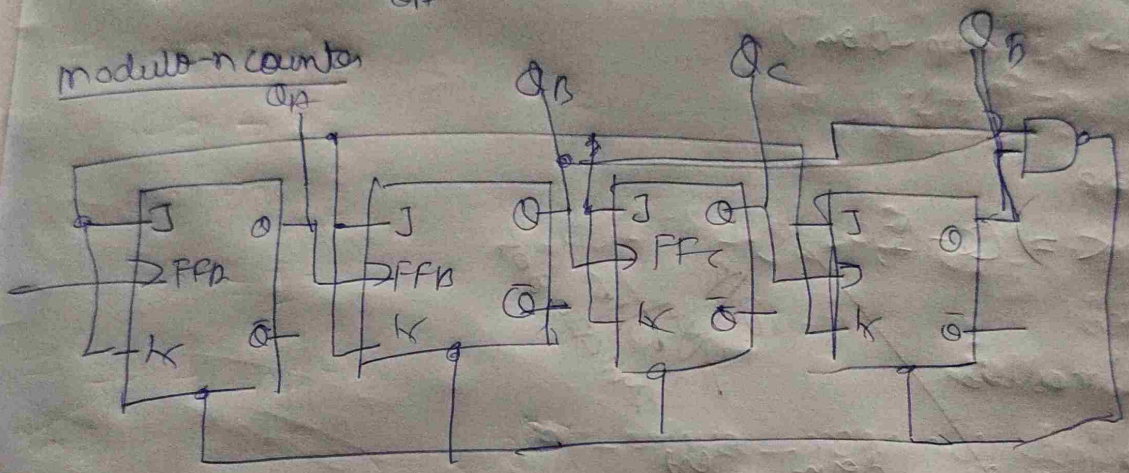
Johnson Counter: a counter where the inverted output of the last flip-flop is connected to the input of the first flip-flop. This creates a sequence of states that is twice the number of flip-flop. for example, a 3-stage Johnson counter has a modulus of 6.

Circuit Diagram

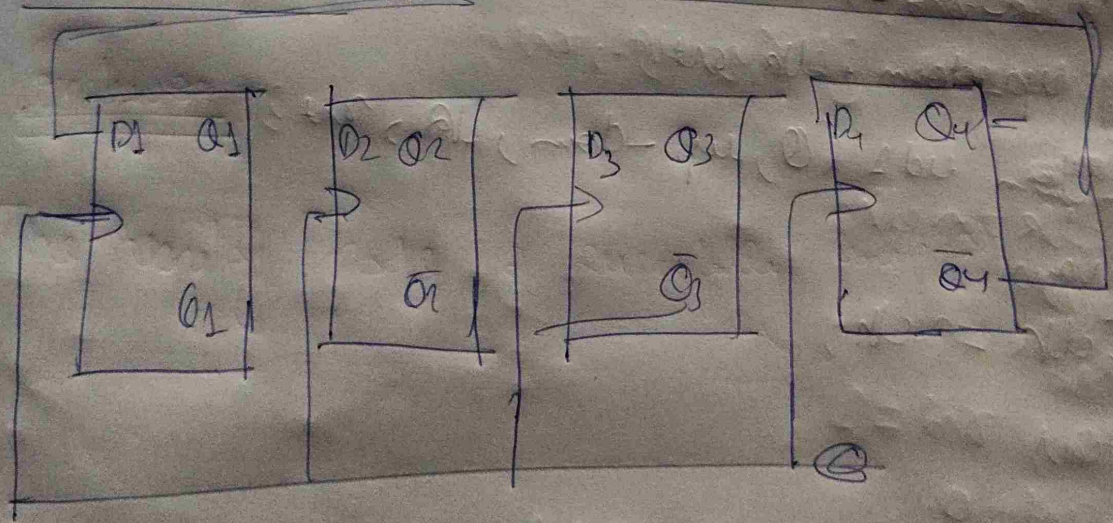
UP-down



modulo-n counter

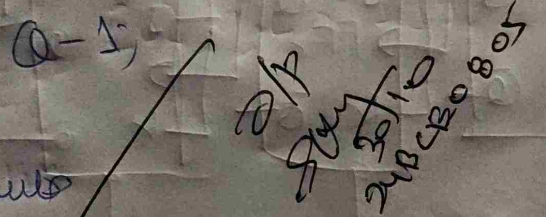


Johnson Counter



Q code

```
module up_down (Q, up_dow, clk, rst);
output [3:0] Q;
input up_dow, clk, rst;
reg [3:0] Q;
always @ (posedg clk or posdeg rst);
begin
if (rst)
Q <= 4'b0;
else if (up_dow)
Q <= Q+1;
else
Q <= Q-1;
end
endmodule
```



```
module up_dow_test;
wire [3:0] Q;
reg up_dow, clk, rst;
up_dow ud1 (Q, up_dow, clk, rst);
always
# 5 clk = ~clk;
initial
begin
clk=0; up_dow =1; rst=1;
# 10 rst=0;
# 100 up_dow =0;
# 200 $stop;
```

```
end
endmodule
```

b) Code

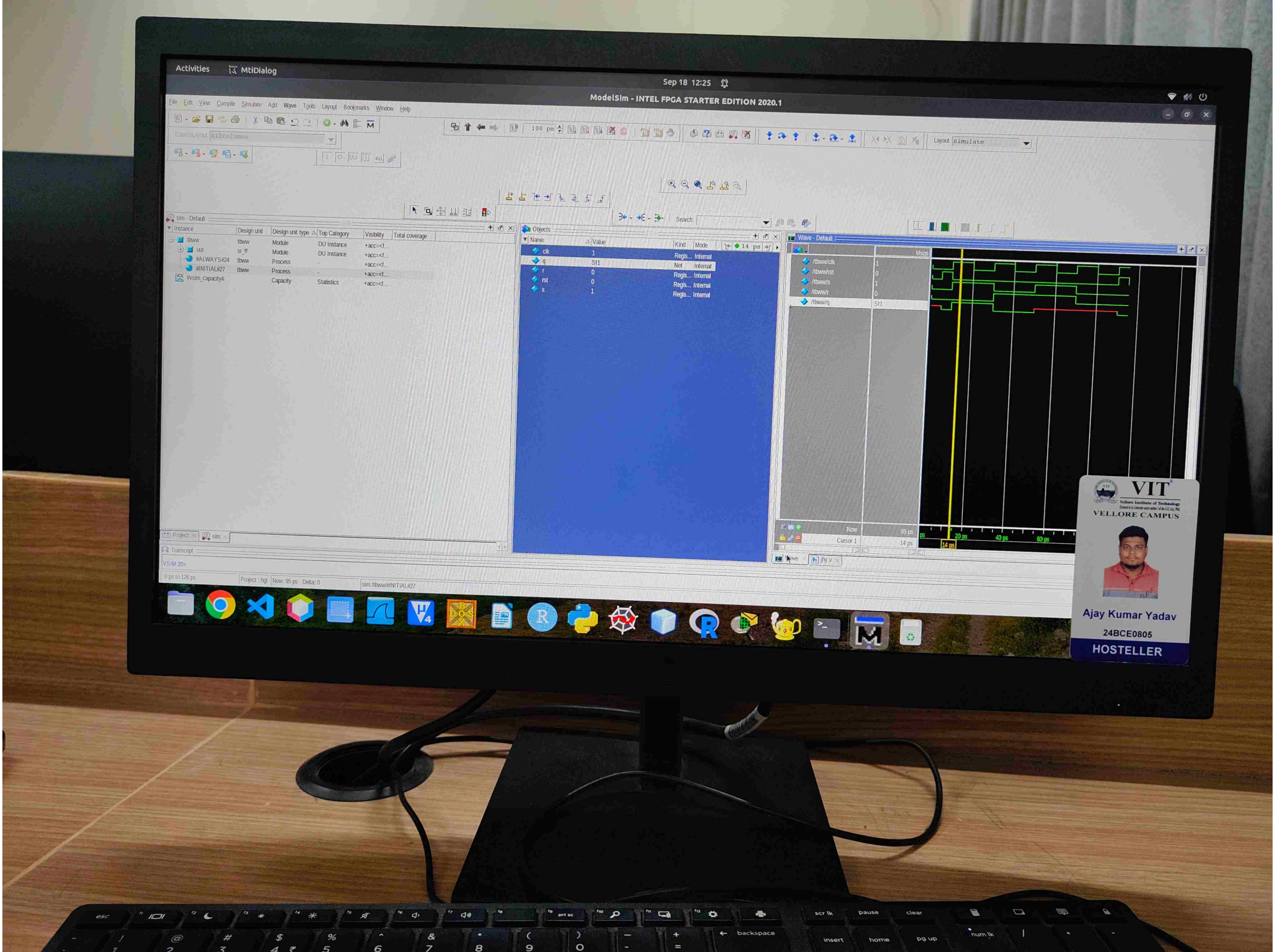
```
module modN_counter (Q, clk, rst);
output [3:0] Q;
input clk, rst;
reg [3:0] Q;
always @(posedge clk or posedge rst);
begin
if (rst)
Q <= 4'b0;
else if (Q==9)
Q <= 4'b0;
else
Q <= Q+1;
end
endmodule

module modN_counter_tb;
wire [3:0] Q;
reg clk, rst;
modN_counter mdl (Q, clk, rst);
always
#5 clk = ~clk; // IP
initial
begin
clk = 0; rst = 1;
#10 rst = 0;
#200 $stop;
```

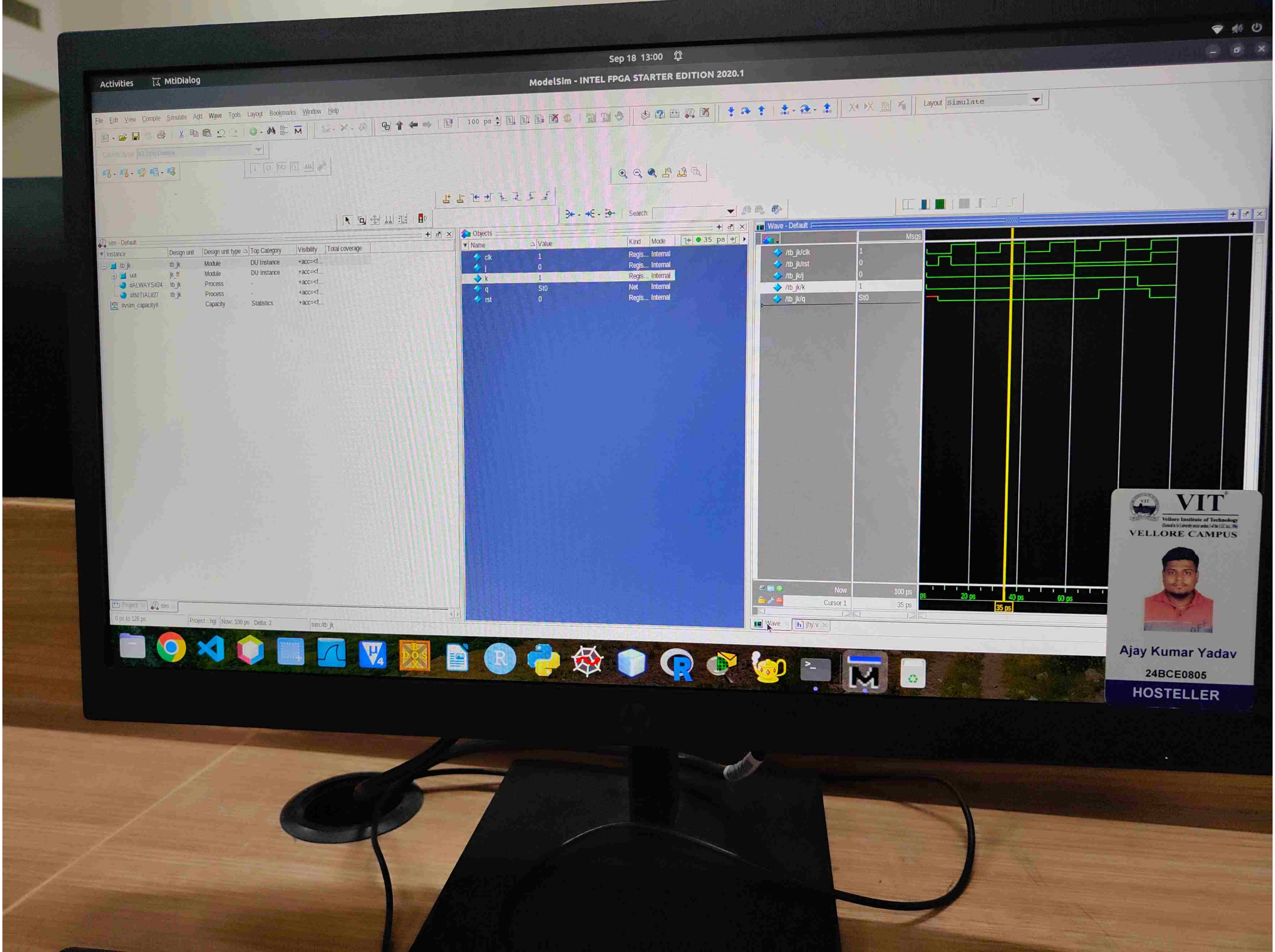
end
endmodule

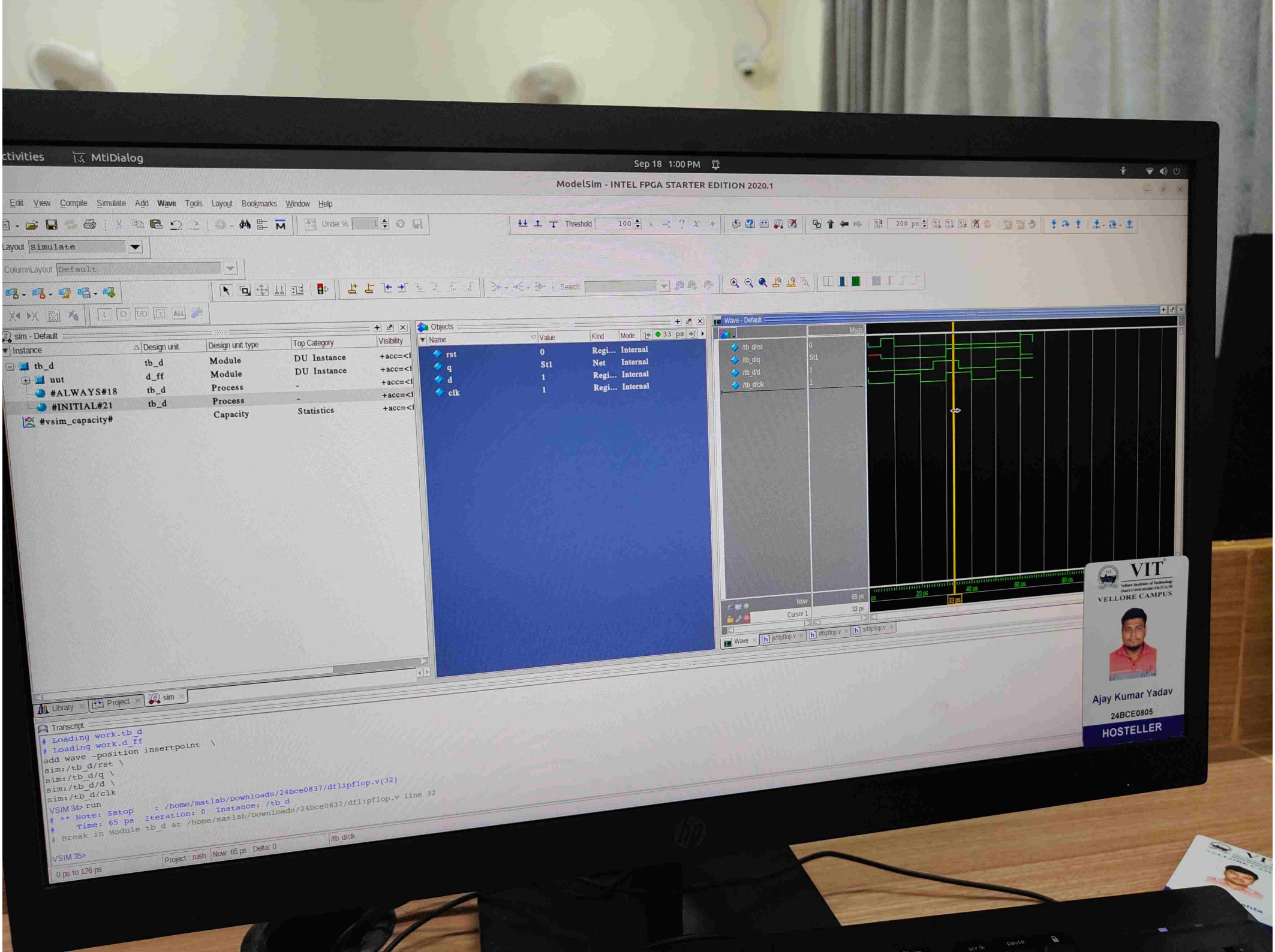
(c) Johnson Counter

```
module johnson_counter (q, clk, rst);
output [3:0] q;
input clk, rst;
reg [3:0] q;
always @(posedge clk or posedge rst);
begin
if (rst)
q <= 4'b0000;
else
q <= {~q[0], q[3:1]};
end
endmodule
module johnson_counter_tb;
wire [3:0] q;
reg clk, rst;
johnson_counter jc (q, clk, rst);
always
#5 clk = ~clk;
initial
  begin
clk = 0; rst = 1;
#10 rst = 0;
#200 $stop;
end
endmodule
```

O/P verified
Dilip
30-10-2025


Activities
MtiDialog
Sep 18 12:25
ModelSim - INTEL FPGA STARTER EDITION 2020.1
VIT
Vellore Institute of Technology
VELLORE CAMPUS
Ajay Kumar Yadav
24BCE0805
HOSTELLER


Sep 18 13:00
Activities
MtiDialog
ModelSim - INTEL FPGA STARTER EDITION 2020.1
Wave - Default
Objects
clk
j
k
q
rst
/tb_jk/clk
/tb_jk/rst
/tb_jk/j
/tb_jk/k
/tb_jk/q
VIT
Vellore Institute of Technology
VELLORE CAMPUS
Ajay Kumar Yadav
24BCE0805
HOSTELLER


ctivities
MtiDialog
Sep 18 1:00 PM
ModelSim - INTEL FPGA STARTER EDITION 2020.1
Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance Design unit Design unit type Top Category Visibility
tb_d tb_d Module DU Instance
uut d_ff Module DU Instance
#ALWAYS#18 tb_d Process
#INITIAL#21 tb_d Process
#vsim_capacity# Capacity Statistics
Objects
Name Value Kind Mode
rst 0 Regi... Internal
q St1 Net Internal
d 1 Regi... Internal
clk 1 Regi... Internal
Wave - Default
/tb_d/rst
/tb_d/q
/tb_d/d
/tb_d/clk
Library Project sim
Transcript
Loading work.tb_d
Loading work.d_ff
add wave -position insertpoint \
sim:/tb_d/rst \
sim:/tb_d/q \
sim:/tb_d/d \
sim:/tb_d/clk
VSIM 34> run
** Note: $stop : /home/matlab/Downloads/24bce0837/dflipflop.v(32)
Time: 65 ps Iteration: 0 Instance: /tb_d
Break in Module tb_d at /home/matlab/Downloads/24bce0837/dflipflop.v line 32
VSIM 35>
Project : rush Now: 65 ps Delta: 0
/tb_d/clk
0 ps to 126 ps
VIT
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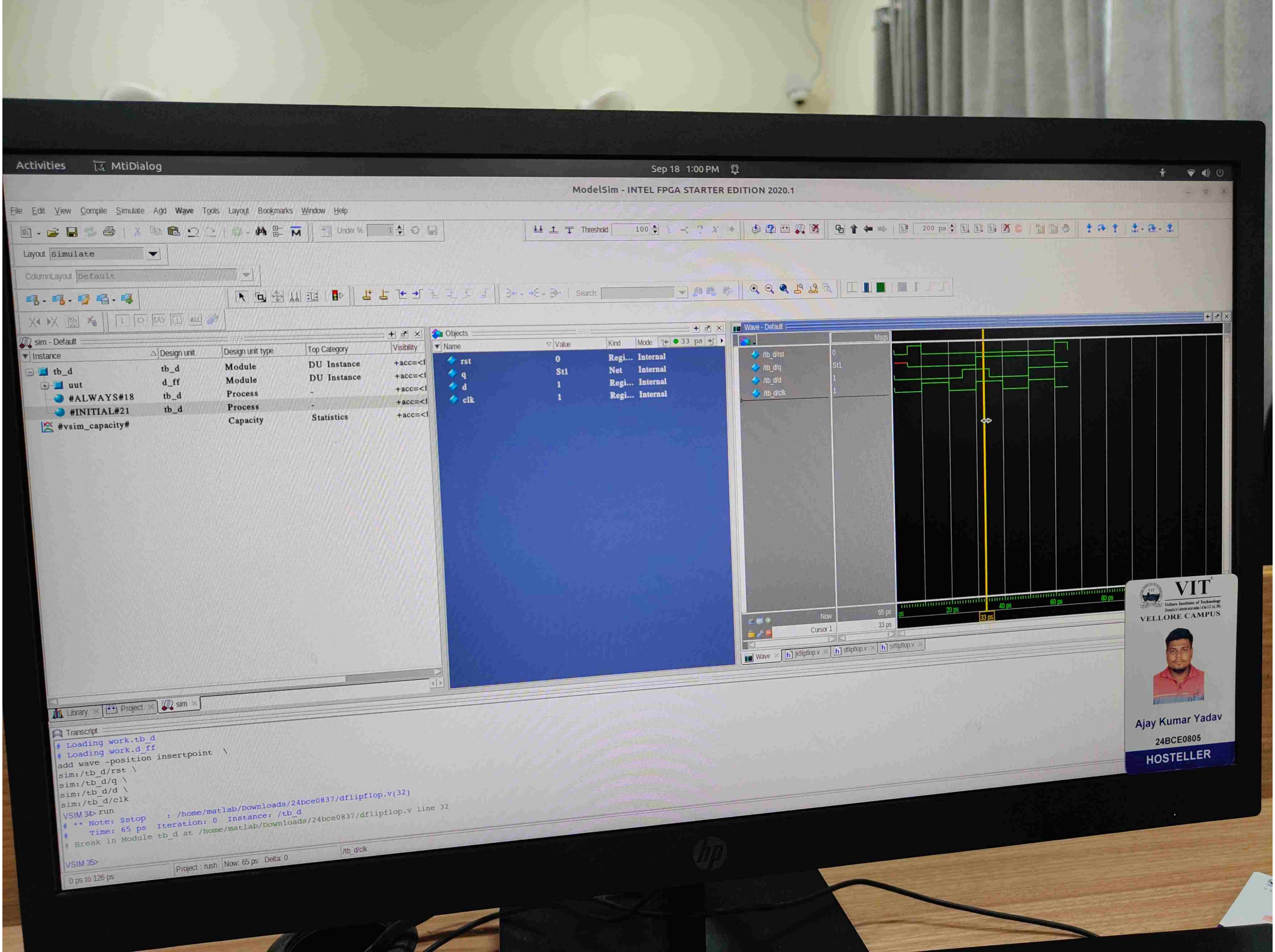
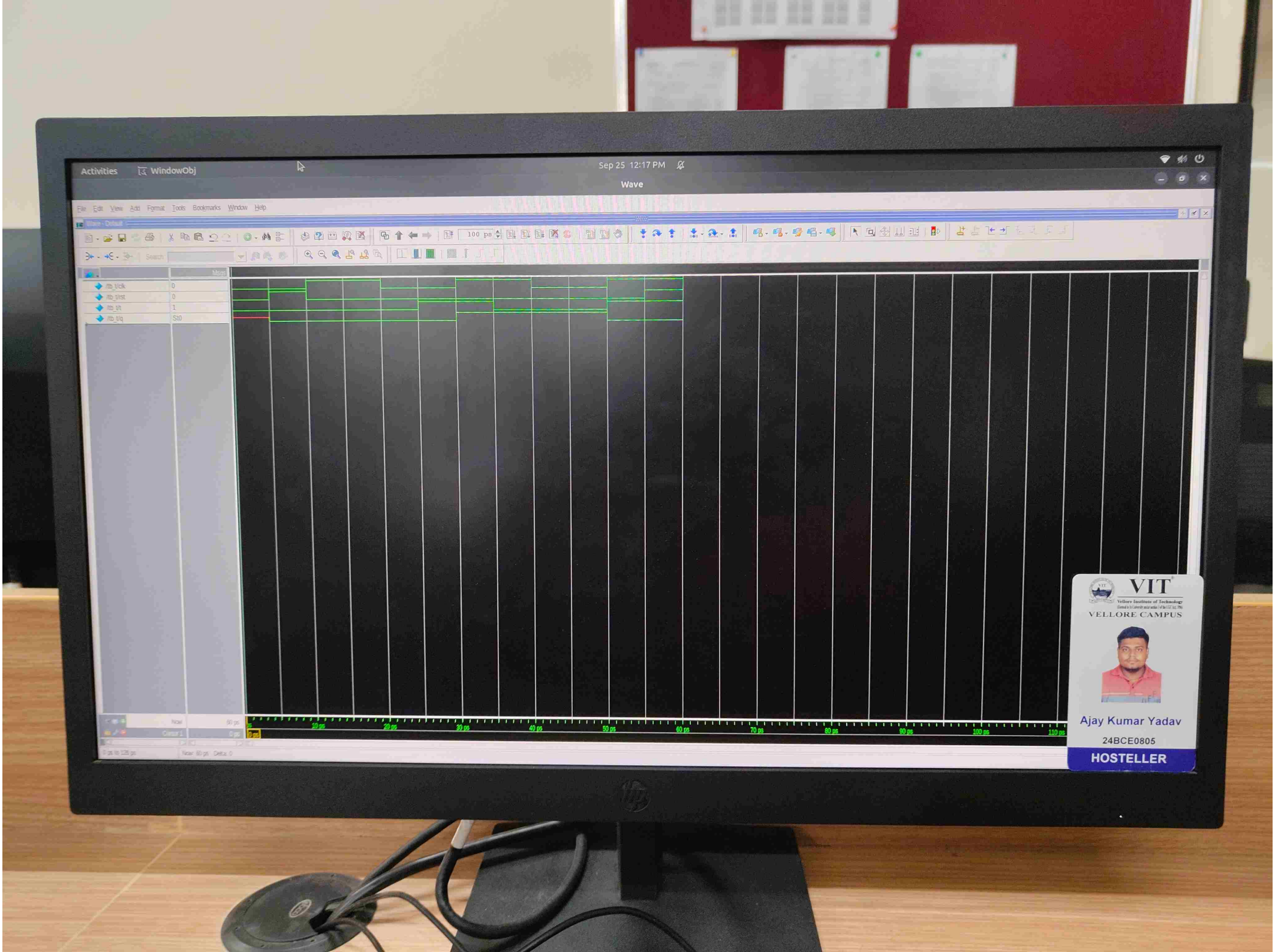
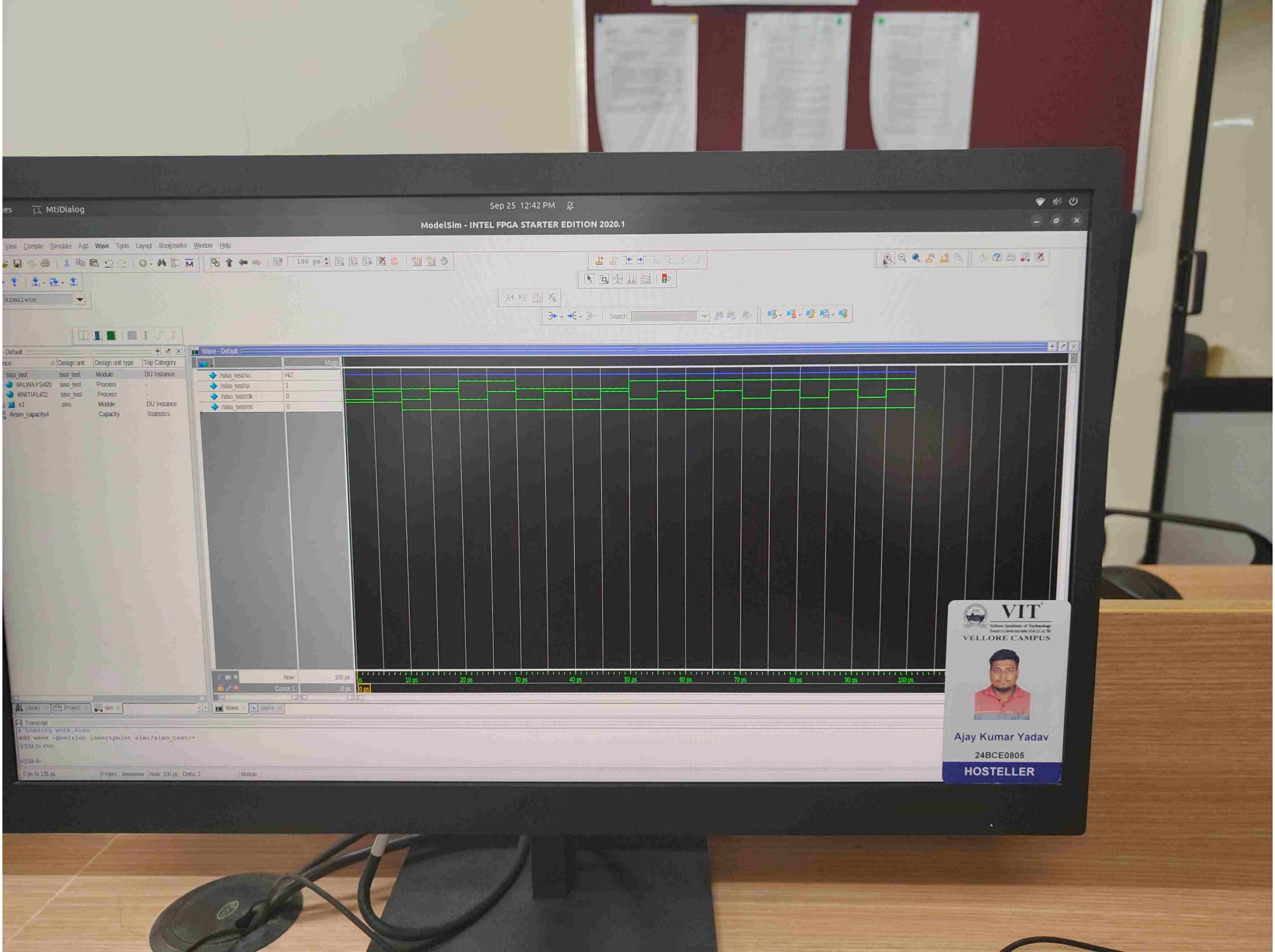
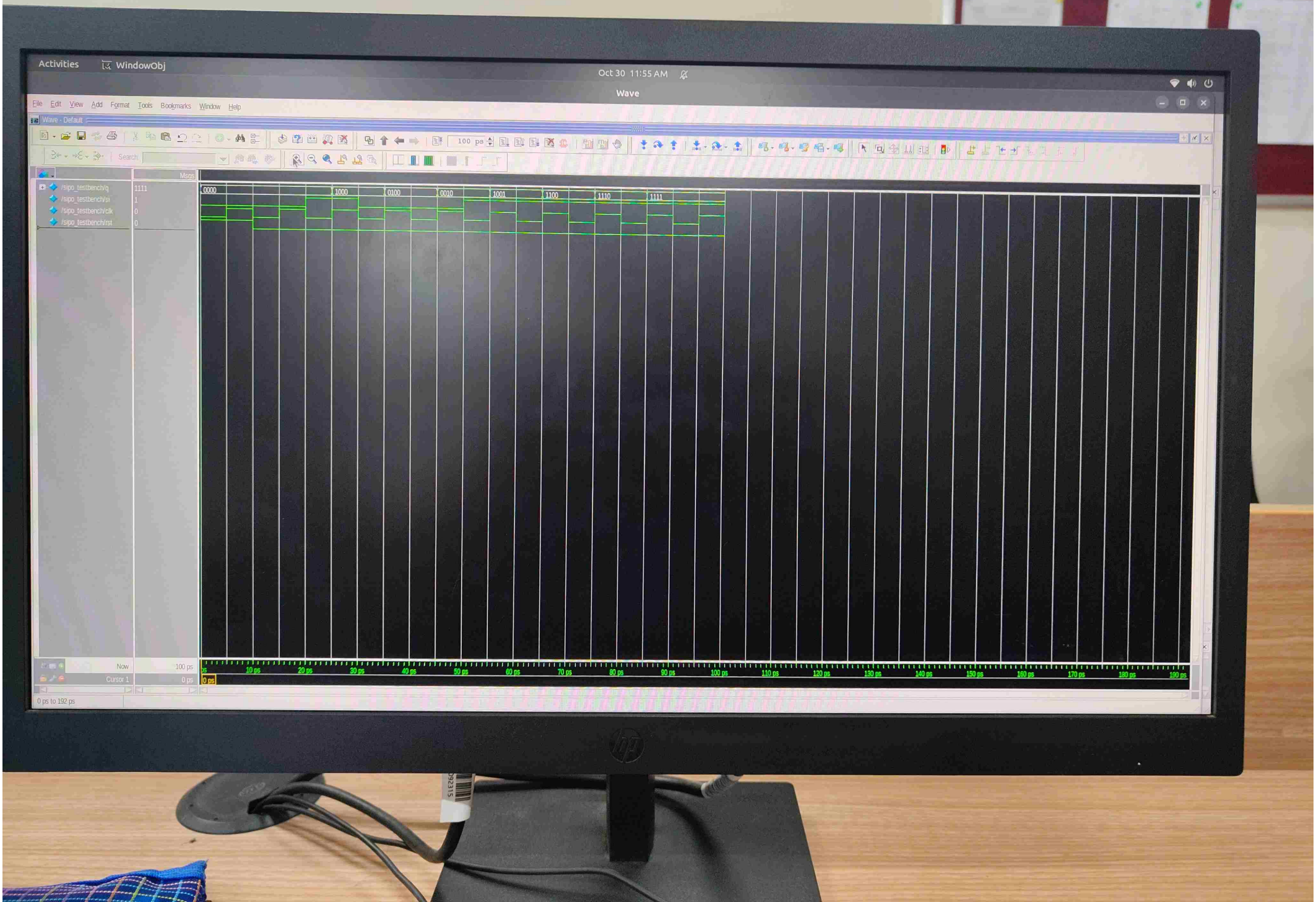
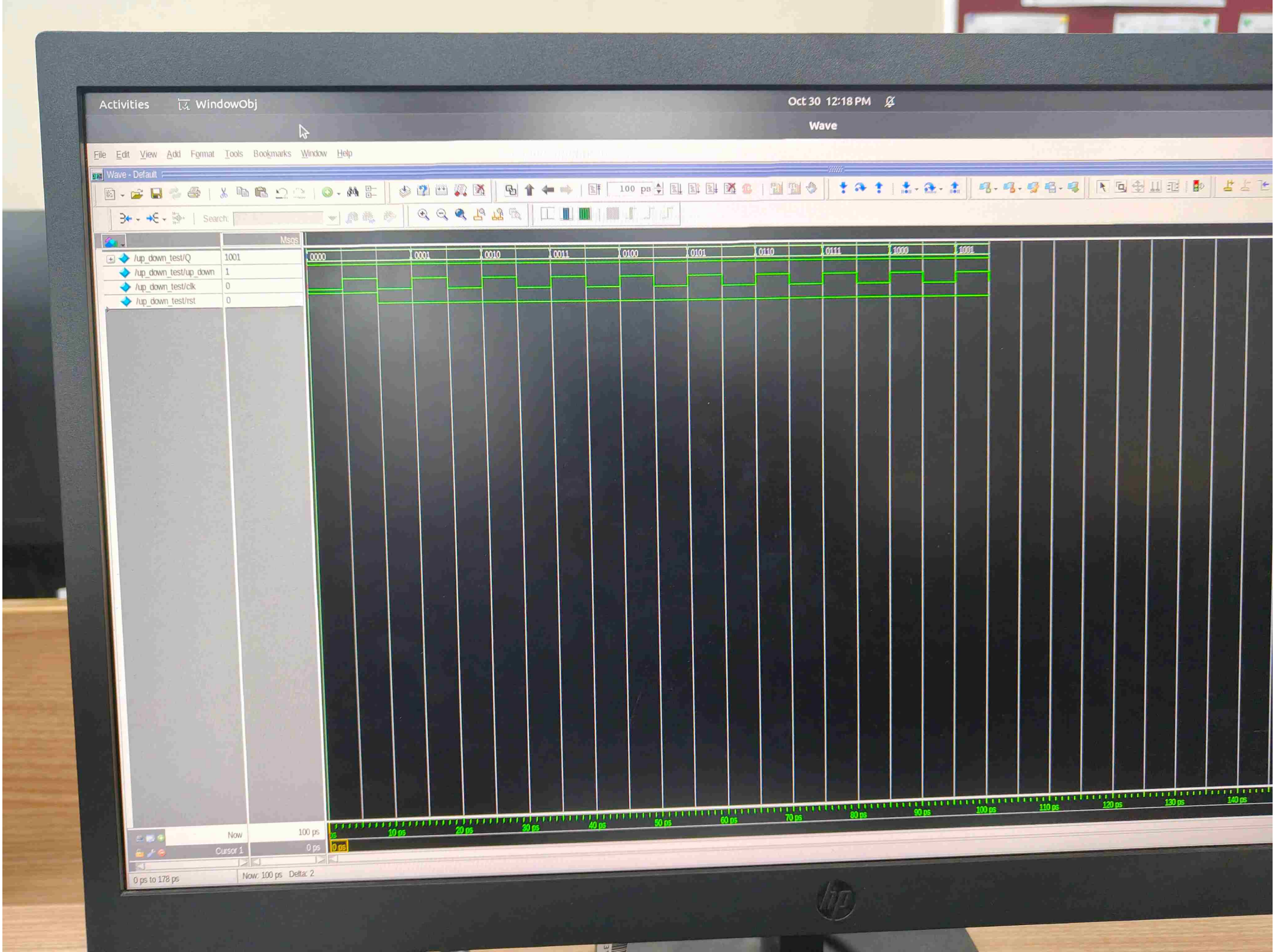
Ajay Kumar Yadav
24BCE0805
HOSTELLER


Activities
MtiDialog
Sep 18 1:00 PM
ModelSim - INTEL FPGA STARTER EDITION 2020.1
File Edit View Compile Simulate Add Wave Tools Layout Bookmarks Window Help
sim - Default
Instance Design unit Design unit type Top Category Visibility
tb_d tb_d Module DU Instance +acc=<
uut d_ff Module DU Instance +acc=<
#ALWAYS#18 tb_d Process - +acc=<
#INITIAL#21 tb_d Process - +acc=<
#vsim_capacity# Capacity Statistics +acc=<
Objects
Name Value Kind Mode
rst 0 Regi... Internal
q St1 Net Internal
d 1 Regi... Internal
clk 1 Regi... Internal
Wave - Default
/tb_d/rst
/tb_d/q
/tb_d/d
/tb_d/clk
Library Project sim
Transcript
Loading work.tb_d
Loading work.d_ff
add wave -position insertpoint \
sim:/tb_d/rst \
sim:/tb_d/q \
sim:/tb_d/d \
sim:/tb_d/clk
VSIM 34> run
** Note: $stop : /home/matlab/Downloads/24bce0837/dflipflop.v(32)
Time: 65 ps Iteration: 0 Instance: /tb_d
Break in Module tb_d at /home/matlab/Downloads/24bce0837/dflipflop.v line 32
VSIM 35>
Project : rush Now: 65 ps Delta: 0
/tb_d/clk
0 ps to 126 ps
VIT
Vellore Institute of Technology
VELLORE CAMPUS
Ajay Kumar Yadav
24BCE0805
HOSTELLER

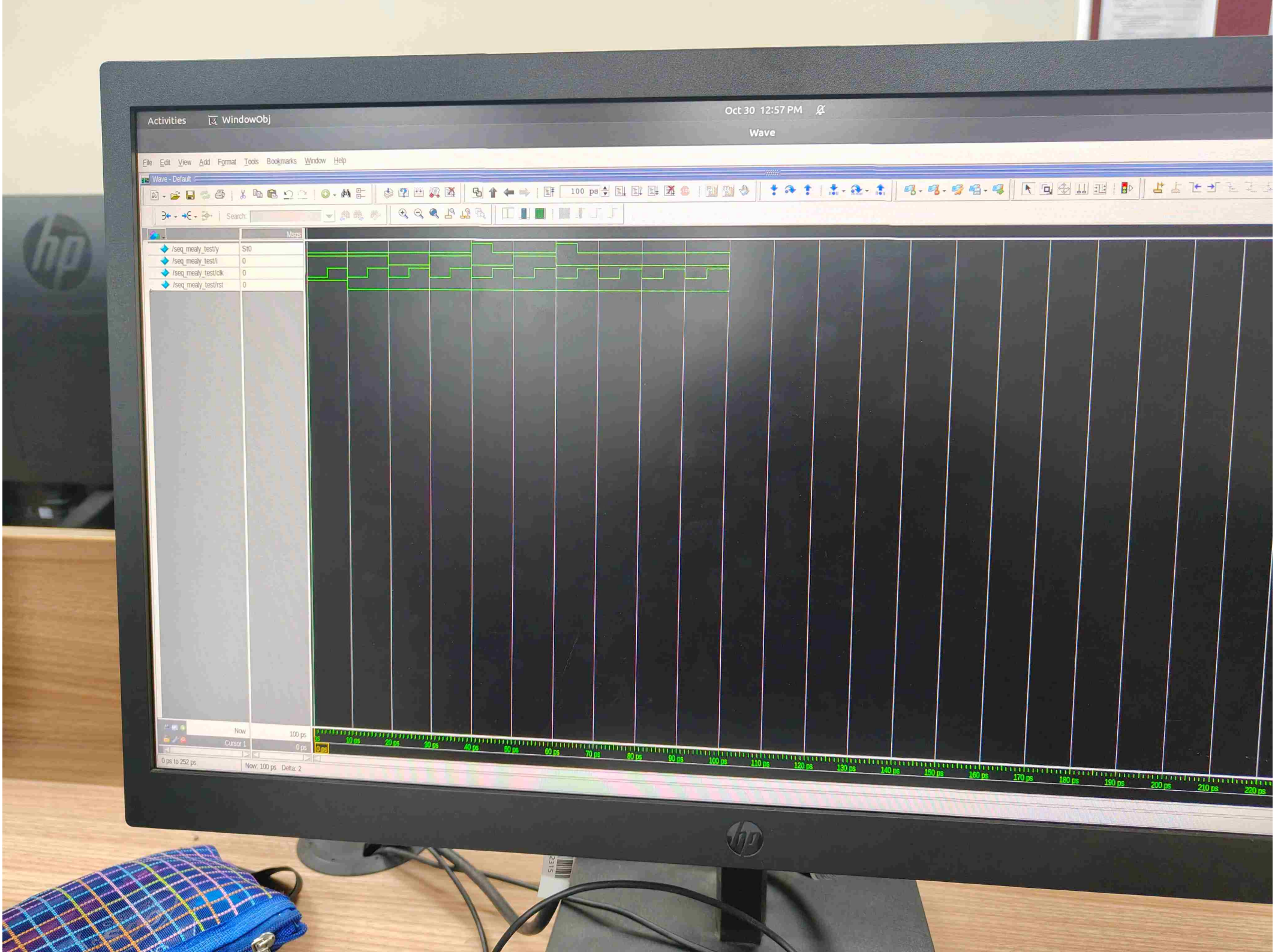

Activities
WindowObj
Sep 25 12:17 PM
Wave
VIT
Vellore Institute of Technology
VELLORE CAMPUS
Ajay Kumar Yadav
24BCE0805
HOSTELLER


MtiDialog
Sep 25 12:42 PM
ModelSim - INTEL FPGA STARTER EDITION 2020.1
Wave - Default
/siso_test/so
/siso_test/si
/siso_test/clk
/siso_test/rst
VIT
VELLORE CAMPUS
Ajay Kumar Yadav
24BCE0805
HOSTELLER


Activities
WindowObj
Oct 30 11:55 AM
Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search
Msgs
/sipo_testbench/q
/sipo_testbench/si
/sipo_testbench/clk
/sipo_testbench/rst
1111
1
0
0
0000
1000
0100
0010
1001
1100
1110
1111
Now
100 ps
Cursor 1
0 ps
0 ps to 192 ps
10 ps
20 ps
30 ps
40 ps
50 ps
60 ps
70 ps
80 ps
90 ps
100 ps
110 ps
120 ps
130 ps
140 ps
150 ps
160 ps
170 ps
180 ps
190 ps


Activities
WindowObj
Oct 30 12:18 PM
Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
/up_down_test/Q
/up_down_test/up_down
/up_down_test/clk
/up_down_test/rst
1001
1
0
0
0000 0001 0010 0011 0100 0101 0110 0111 1000 1001
Now
Cursor 1
100 ps
0 ps
0 ps to 178 ps
Now: 100 ps Delta: 2


Activities
WindowObj
Oct 30 12:36 PM
Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
/johnson_counter_test/q
/johnson_counter_test/clk
/johnson_counter_test/rst
1000
0
0
Msgs
0000
1000
1100
1110
1111
0111
0011
0001
0000
1000
Now
100 ps
Cursor 1
0 ps
0 ps to 126 ps
Now: 100 ps Delta: 2
10 ps
20 ps
30 ps
40 ps
50 ps
60 ps
70 ps
80 ps
90 ps
100 ps
110 ps
120 ps
hp
esc
scr lk
pause


Activities
WindowObj
Oct 30 12:57 PM
Wave
File Edit View Add Format Tools Bookmarks Window Help
Wave - Default
Search:
Msgs
/seq_mealy_test/y
/seq_mealy_test/i
/seq_mealy_test/clk
/seq_mealy_test/rst
St0
0
0
0
Now
Cursor 1
100 ps
0 ps
0 ps to 252 ps
Now: 100 ps Delta: 2
10 ps 20 ps 30 ps 40 ps 50 ps 60 ps 70 ps 80 ps 90 ps 100 ps 110 ps 120 ps 130 ps 140 ps 150 ps 160 ps 170 ps 180 ps 190 ps 200 ps 210 ps 220 ps